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The Pain of Exercise – Insights in to The Role of Pain on Exercise Regulation and Performance

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'No pain, no gain,' 'The man who can drive himself further once the effort gets painful is the man who will win,' 'Pain is temporary, quitting lasts forever'. There is no doubt that when exercise is performed above certain intensity, it elicits feelings of discomfort and pain.

These feelings have been shown to cause negative 'affect' [1], which can reduce adherence to exercise routines and may impact on exercise performance [2]. Thus, further understanding of the physiological mechanisms that elicit these negative feelings associated with exercise-induced pain, may help improve exercise strategies for public health, and increase athletic performance.

The International Association for the Study of Pain has defined pain as an unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage (International Association for the Study of Pain). This definition implies that:

1. Pain is always a subjective experience
2. Emotions are always an element of pain
3. Perception of pain is not always directly related to the level of tissue damage

Type III and Type IV nociceptors, which send peripheral information to the central nervous system (CNS) [3], are stimulated by mechanical pressure, heat, cold, high noxious pressure and endogenous pain producing (algesic) substances such as bradykinin, potassium, serotonin, histamine, hydrogen ions and prostaglandins. Intense skeletal muscle activity leads to an increase in these substances and contributes to the acute muscle pain associated with particular forms of exercise [4]. Increased levels of pain have been shown to reduce voluntary muscle contraction and increase avoidance behaviour [1], yet little research exists on the mechanisms underpinning pain associated with exercise.

Fatigue can be defined as any exercise-induced reduction in the ability to exert muscle force or power, regardless of whether or not the task can be sustained [5]. Whilst traditionally, this has been explained from a peripheral standpoint [6], recently more focus has been placed on central mechanisms of fatigue [5,7]. Recent theories of central control [7] suggest that afferent feedback from the periphery is collated and processed in conjunction with past experience and current knowledge of the external environment, to produce a 'sensation' of fatigue on which the level of motor output is based. If this is the case, then exercise-induced pain may be a likely contender contributing to the sensation of fatigue.

Despite this link, limited research has been conducted on the fatiguing effects of exercise-induced pain. Cycle ergometry has consistently been documented as causing naturally occurring muscle pain [4], with one study [8] showing that the magnitude of leg muscle pain increases as a positively accelerating function of the percentage of peak power output. Furthermore, when pain is increased during exercise, through the administration of an opioid receptor antagonist such as naloxone, performance is reduced in the presence of greater levels of pain [9]. Thus, it appears as though artificially increased pain during exercise leads to faster fatigue for the same level of intensity.

Only a few recent studies have investigated the effects of reducing the pain response during exercise. Amann et al. [10] found that the injection of intrathecal fentanyl (a peripheral analgesic) to impair cortical projection of opioid-mediated muscle afferents allowed cyclists to endure a higher development of peripheral muscle fatigue. Interestingly the authors reported that the participants had severe problems dismounting the ergometer following testing, and reported ambulatory problems in the hours following testing. Thus, it may be that the perception of exercise induced pain serves some protective, over-exertional function. However, fentanyl reduces the exercise pressor reflex, and the migration of fentanyl from the area of injection may have been responsible for the observed effects in Amann's study. A controlled reduction of pain during exercise is problematic, as many analgesics elicit physiological or psychological responses in addition to analgesia. Acetaminophen, commonly known as paracetamol, can easily be administered in a double-blind fashion and other than analgesia and febrifuge, has no other known effects that may effect exercise performance. Mauger et al. [2] found that this centrally-acting analgesic allowed cyclists to maintain a higher power output, under higher metabolic cost, for the same level of perceived pain and exertion compared to a placebo controlled group. This was associated with an increased performance that was not seen in the study by Amann et al. [10], indicating that the sensation of exercise-induced pain in these two studies was perceived very differently.

The role exercise-induced pain as a contributor to fatigue and exercise tolerance also has implications for clinical populations. A common symptom of sufferers of chronic pain, chronic fatigue syndrome, fibromyalgia, depression and multiple sclerosis is that exercise causes pain during the activity and a prolonged period of post-exertional malaise after exercise [11]. The literature is equivocal as to whether the increased pain response in these conditions is due to heightened peripheral nociceptive sensitivity or to an increased central processing of the same sensory input [12]. Furthermore, negative affect during exercise (which has likely associations with discomfort or pain) has been shown to reduce exercise adherence [1] and this may have further implications for clinical populations using exercise for health related reasons.

Given the long and well-established link between exercise and pain, it is curious that comparatively few studies have focused on this area. However, the potential impact for public health and exercise performance that the increased understanding of this relationship may

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yield is significant. Therefore, future work should look to examining the role of exercise-induced pain on the tolerance, performance and regulation of exercise, both in the performance athlete, the 'normal' individual, and the clinical patient - 'Your pain is the breaking of the shell that encloses your understanding'.

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